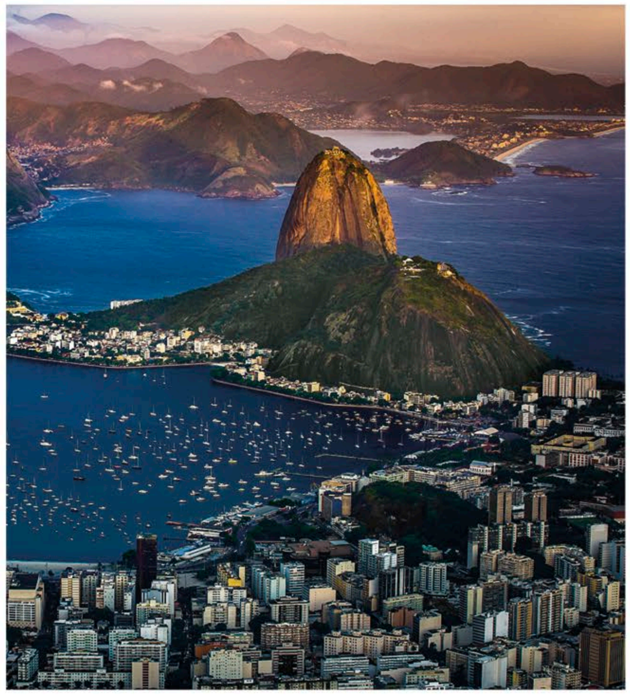
The Sugarloaf Mountain

A monolith made from granite and quartz that rises vertically from the water's edge in Rio de Janeiro

Rising to a height of 1,299 ft (396 m) above Rio de Janeiro, the Sugar Loaf is situated on a peninsula that juts out into the mouth of Guanabara Bay—a large sea inlet that includes Rio's harbor.

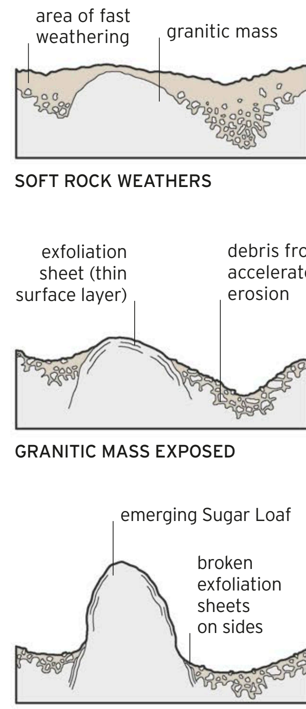
The rock from which the Sugar Loaf and other nearby dome-shaped mountains are composed was formed millions of years ago from magma rising from deep underground and then solidifying beneath the surface. Starting around 100 million years ago, South America separated from Africa to form the southern Atlantic Ocean, resulting in fractures in Earth's crust around the eastern coast of South America. The fractures created weaknesses in the rocks surrounding the granitic masses, allowing them to be broken down by agents such as rain (see panel, right).



▷ **RISING STRAIGHT UP**
From the top of the Sugarloaf (in the center in this image), there is a bird's-eye view of both Rio's harbor, seen in the foreground here, and Guanabara Bay, visible on the mountain's left.

HOW THE SUGAR LOAF FORMED

The Sugar Loaf began as a mass of granite underground. It was exposed at the surface as the softer rock above and around it was weathered away. Further erosion created its shape.



The Hornocal Mountains

A small mountain range in northwest Argentina with a spectacular zigzag structure and incredible colors

▽ **DELUGE OF COLOR**
The hues of the Hornocal Mountains' rock layers range from vivid reds to creams and gray-greens.

The Hornocal Mountains are part of a formation of sedimentary rocks that extends for some 310 miles (500 km) southward from Peru. Known as the Yacoraite formation, it is composed mainly of limestones, as well as some sandstones and siltstones, and was formed in a shallow sea between 72 and 61 million years ago (when this part of South America was underwater).

Variations in the mineral composition of the sediments deposited into the sea, the chemistry of the seawater, and the life-forms present at the time account for the striking variation in the rock colors. Over the years, these rock layers were lifted up and tilted as a result of tectonic plate movements, then weathered and eroded, resulting in the breathtaking sight that is visible today. The riot of colors lends a slightly surreal appearance to the location.